

What is a Hardware Firewall?

Extremely large networks have what is known as a Hardware Firewall. This is a group of dedicated servers running a special OS and nothing else but the firewall. These servers are placed along the periphery of the network, such that no access to the Internet is available except through them. This creates a demilitarised zone (DMZ), an area that cannot be trespassed by unauthorised individuals.

the results to its server. It also did a terrible job at stopping Port Fun from opening up ports on the test bed.

Price: \$39.95 ■ **System requirements:** Windows XP, 2000, NT, Me, 98, 95, Pentium 133 or equivalent CPU, 32 MB RAM, 10 MB free disk space, IE 5.0 or later ■ **Web site:** www.sygate.com ■ **Grade:** A-

Microsoft XP Internet Connection Firewall:

The default Internet Connection Firewall (ICF), bundled with Windows XP, is as barebones as can be. It got the maximum score in installation since you just have to click on a check box to enable the firewall. It was also very easy to configure, although the number of features that could be configured was pretty much less than any other firewall we tested. But it did a good job at resisting port scans. One good thing about the ICF is that it has a very high default security level, although e-mail clients such as Outlook Express can still access and download e-mails unrestricted. Another shortcoming is the absence of alerting mechanisms and a very cryptic log file that hardly makes any sense to the layman.

Price: Free with Windows XP ■ **System requirements:** Bundled with Windows XP ■ **Web site:** www.microsoft.com ■ **Grade:** B

Kerio Personal Firewall 2.1.5:

This firewall from Kerio Technologies stood fifth overall—not really impressive, but it does have some features that were absent in the other firewalls. For one, it has a feature that saves configuration settings in multiple config files, so that they can be loaded as and when required—you don't have to reconfigure tasks such as letting your colleagues access your shared folders. Unfortunately, it could not stop Port Fun from opening up ports, even when set to the highest security level.

Price: Free for personal use; \$39 for a single user commercial license ■ **System requirements:** Pentium class Processor or higher; Windows 9x, NT4, ME, 2000, XP; 32 MB RAM; 10 MB available disk space ■ **Web site:** www.kerio.com ■ **Grade:** B+

The Honeypot project

A Honeypot is a machine that is put up on the Internet with the purpose of attracting hackers and then extensively tracking their behaviour. The results obtained from thousands of such Honeypots, spread across the Internet, are then analysed by the Honeypot project researchers and released in the form of reports or books that are used to develop effective counter attacks.

Norton Personal Firewall 2003:

This is a very efficient firewall that resisted all port scan attempts. Norton Personal Firewall uses the same LiveUpdate utility as Norton AntiVirus. It offered excellent resistance to port scanning. In fact, it even stopped Open Ports from running the netstat command and uploading results to the server. It has a friendly security assistant that lets you configure home networking, program scans, privacy control and password protection.

Price: \$49.95 ■ **System requirements:** Windows XP, Pentium II 300 MHz or higher processor, Windows 2000 Pro, Me, 98, Intel Pentium 150 MHz, 25 MB hard disk space, 64 MB RAM, IE 5.01 ■ **Web site:** www.symantec.com ■ **Grade:** A

ZoneLabs ZoneAlarm Pro v3.7.159:



ZoneAlarm has been one of the best firewalls for quite some time; and it was no different this time. It was easy to install, configure and provided very good protection.

Immediately after the install, the configuration wizard guides you through the privacy control, level of alerting and password protection settings. You can also let ZoneAlarm preconfigure permissions for the browser and its components to access the Internet. After this, you can walk through a nine-step tutorial that will get you started with configuring the firewall.

The MailSafe feature lets you quarantine, or allow attachments based on their file types. For all its bells and whistles, ZoneAlarm does place some serious overhead on your Internet browser.

Price: \$40 ■ **System requirements:** Windows 98 SE, Me, 2000 Pro, XP; 233 MHz Pentium CPU or higher; 10 MB of available hard disk space, Internet access, 48 MB for 98 SE/Me, 64 MB RAM for 2000 Pro, 128 MB RAM for XP ■ **Web site:** www.zonelabs.com ■ **Grade:** A+

ISS BlackICE PC Protection 3.6:

This firewall profiles your computer to create a baseline of all the existing applications. It identifies and defines nmap scans, giving limited information about what it does in a small event window. It also identified the nmap OS fingerprinting option, and fooled the port scanner by sending out many fingerprints. However, it could not stop Open Ports from identifying and uploading results to its server, nor could it stop Port Fun.

BlackICE also provides back-tracing of the intruders using DNS lookups and NetBIOS node status. While it was able to trace all information about the invading Windows machine, it could only find the IP of the Linux hacker. It gives visual and auditory event notification. It could not terminate an already running Port Fun, but did terminate the instance that was invoked later. However, the blocking features were dismal, since it could not block Port Fun from opening up ports (although it said that the action had been blocked in logs), once the termination condition was removed.

Price: \$39.95 ■ **System requirements:** Windows 98, ME, NT 4.0, 2000, XP Pro, XP Home, Pentium class CPU or better, 16 MB RAM, 10 MB free disk space ■ **Web site:** www.iss.net ■ **Grade:** B+